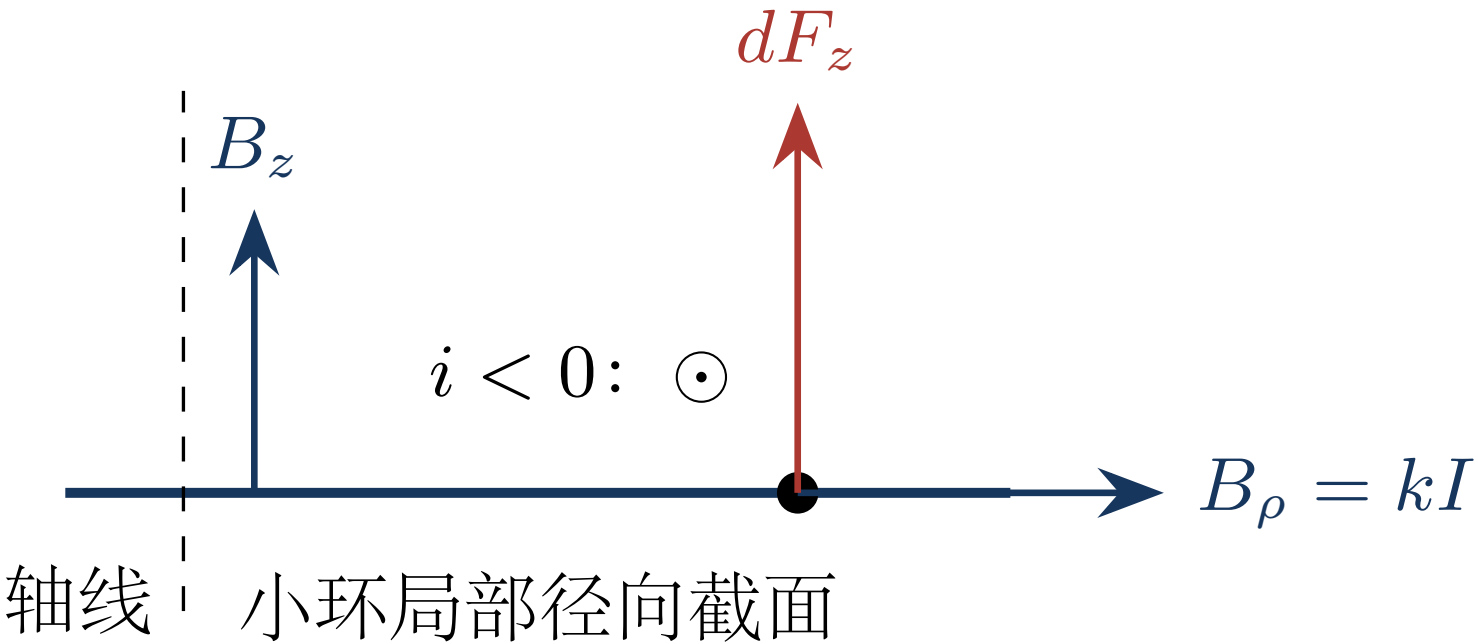

$$F_z = -2\pi RkIi$$